

Loss in Low-Finesse Ti:LiNbO₃ Optical Waveguide Resonators

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Abstract. We propose to measure the contrast of Fabry-Perot resonances in low-finesse (end face polished) integrated optical resonators to determine an upper limit of the waveguide attenuation coefficient. The method is discussed; its absolute accuracy increases with decreasing loss. As example, experimental results are presented for 3 and 10 μm wide Ti:LiNbO₃ strip guides; the attenuation coefficients are 0.46 dB/cm (EH₁₁) in the narrower guide at $\lambda=0.63 \mu\text{m}$, respectively 0.19 dB/cm (EH₁₁) and 0.09 dB/cm (HE₁₁) in the 10 μm wide guide at $\lambda=1.15 \mu\text{m}$.

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Optical waveguides are the basic components of integrated optical modulators, switches, filters, sensors and nonlinear optical frequency converters. The performance of all these devices, especially of the resonant structures, depends on the waveguide attenuation.

Therefore, a number of methods has been developed in the last years to determine the waveguide losses: controlled prism coupling into and out of a waveguide can yield the attenuation coefficient [1]. To become independent of the coupling efficiency of the input coupler a sliding output coupler has been used [2]. In pyroelectric materials like LiNbO₃ the pyroelectric effect has been exploited to measure the total loss by absorption of the guided light in a mercury drop on top of the waveguide [3]. It was further possible to measure the loss due to the intrinsic material absorption alone by a measurement of a pyroelectric current [4]. Both methods are, however, restricted to pyroelectric materials and certain crystal orientations. By calorimetric methods waveguide absorption and total losses could be determined [5, 6]. Furthermore, the position dependent intensity of the scattered light was analyzed to obtain the waveguide losses [7, 8]. Kaminow and Stulz [9] were the first to use a waveguide Fabry-Perot structure to determine the loss of a nominal 4 μm wide strip guide. They used cleaved LiNbO₃ substrates to get perfect and parallel waveguide end faces. However, there is only one cleavage

plane in LiNbO₃; it forms an angle of about 33° with the optic axis of the crystal. This results in a special waveguide orientation with interference effects for the ordinary wave only.

We demonstrate in this paper that it is possible to use low-finesse waveguide Fabry-Perot resonators in standard crystal orientations to evaluate the total loss of both polarizations. The measurement of the contrast of the resonances is sufficient to calculate a combined loss-reflection factor and thus an upper limit of the attenuation coefficient. It is not necessary to measure absolute loss figures nor to cut back the crystal as was done by Kaminow and Stulz [9]. We discuss the accuracy of the method. It is necessary to use single-frequency lasers or to correct the measured interference contrast dependent on the intensity ratios of the different longitudinal laser modes. Finally, we present some experimental results. The resonator method proves to be especially valuable for characterizing samples which are to be used for a fabrication of high-finesse waveguide resonators (e.g., optical spectrum analyzer, resonant second harmonic generator, optical parametric oscillator [10]).

1. Experimental Apparatus

The schematical experimental arrangement is shown in Fig. 1. We used a polarized single-frequency He-Ne

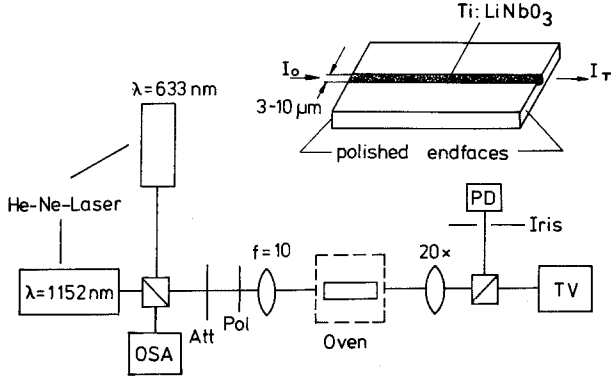


Fig. 1. Schematic experimental setup. (OSA: optical spectrum analyzer; Att: attenuator, Pol: polarizer; TV: television system; PD: photodiode)

laser ($\lambda = 0.63 \mu\text{m}$), respectively a polarized multimode He-Ne laser ($\lambda = 1.15 \mu\text{m}$) as light source. Its spectrum could be measured by an optical spectrum analyzer. The laser beam was end-fire coupled to the low-finesse strip waveguide resonator, placed in a small crystal oven. The mode pattern of the transmitted light was magnified and could be observed by a TV system. Simultaneously, a fraction of the transmitted intensity could be measured by a photodiode as function of the crystal temperature.

The waveguide resonators were fabricated by the standard Ti-indiffusion technique and by carefully polishing the waveguide end faces perpendicular to the waveguide axis. Details of the fabrication process are described elsewhere.

2. Evaluation of Waveguide Loss

The transmitted intensity I_T of a symmetrical mono-mode Fabry-Perot waveguide resonator is given by the standard formula:

$$I_T = \frac{T^2 \exp(-\alpha L)}{(1 - \tilde{R})^2 + 4\tilde{R} \sin^2(\phi/2)} I_0 \eta, \quad (1)$$

where I_0 is the laser beam intensity in front of the waveguide, η the coupling efficiency to the (fundamental) waveguide mode, T the end-face mode transmissivity, and $\phi = 2\beta L$ the internal phase difference with the propagation constant β of the investigated mode in a crystal of length L . The quantity $\tilde{R} = R \exp(-\alpha L)$ is determined by analyzing the waveguide resonances as described below. R is the mode reflectivity and α the attenuation coefficient of the optical waveguide.

The transmitted intensity varies periodically with the optical phase difference ϕ (Fig. 2), which can be tuned by the temperature (or by an applied electrical field). The attenuation coefficient can be evaluated by measuring the absolute values of the maximum and

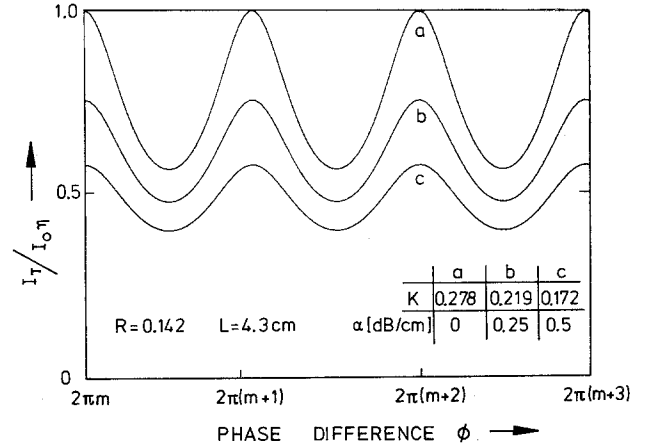


Fig. 2. Calculated resonator transmission versus phase difference. Parameters are resonator length L , end-face reflectivity R and attenuation coefficients α (see inset). K is the corresponding contrast

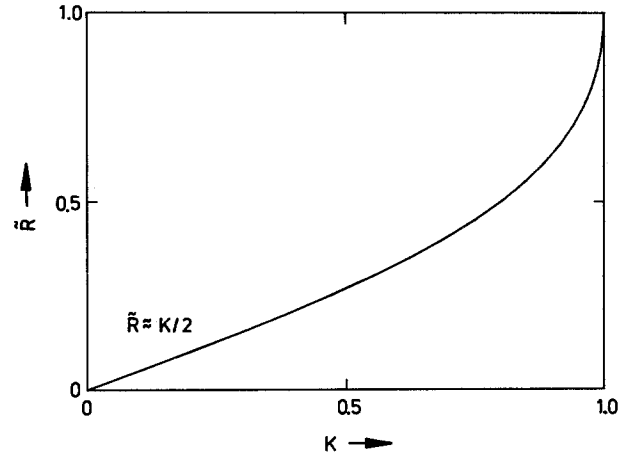


Fig. 3. Combined loss-reflection factor $\tilde{R} = R \exp(-\alpha L)$ as function of the resonances' contrast K

minimum resonator losses [9]. However, a more convenient method is to determine the contrast K of the Fabry-Perot resonances:

$$K = \frac{I_{\max} - I_{\min}}{I_{\max} + I_{\min}}, \quad (2)$$

where K is independent of I_0 and η and a function of $\tilde{R}$ alone, i.e.,

$$K = \frac{2\tilde{R}}{1 + \tilde{R}^2}. \quad (3)$$

Therefore, the combined loss-reflection factor $\tilde{R}$ is given by the measured contrast K

$$\tilde{R} = \frac{1}{K} (1 - \sqrt{1 - K^2}). \quad (4)$$

This function is plotted in Fig. 3; for small K , $\tilde{R} \sim K/2$. If the end-face reflectivity is known it allows to

evaluate the attenuation coefficient α . As $\tilde{R} = R \exp(-\alpha L)$, α is given in dB/cm by

$$\alpha = \frac{4.34}{L} (\ln R - \ln \tilde{R}). \quad (5)$$

Following the analysis, as given in [11], the end-face reflectivity R can be approximated for weakly guiding structures as Ti:LiNbO₃ waveguides by

$$R = \left(\frac{n_{\text{eff}} - 1}{n_{\text{eff}} + 1} \right)^2 \quad (6)$$

with $n_{\text{eff}} = \beta/k_0$ and $k_0 = 2\pi/\lambda$. Equation (6) is a good approximation for the reflectivity of perfect waveguide end-faces; otherwise – due to imperfections induced by the polishing procedure – their reflectivity is lower. Formula (5) therefore yields an upper limit of the waveguide attenuation coefficient.

To illustrate the method, we have plotted in Fig. 4 α as function of the contrast K with $R = 0.142$ (based on $n_{\text{eff}} = 2.209$ for the EH₁₁ mode of the example in Fig. 5), $R = 0.1$ and $L = 4.3$ cm as parameters. For small K as in our examples of low-finesse waveguide resonators, $\tilde{R} \sim K/2$, yielding

$$\alpha \sim \frac{4.34}{L} (\ln R + \ln 2 - \ln K), \quad (7a)$$

respectively,

$$K \sim 2R \exp\left(-\frac{L}{4.34} \alpha\right). \quad (7b)$$

It is evident that a maximum contrast is obtained with a lossless waveguide. Increasing (relatively low!) losses reduce the contrast K drastically, allowing to determine (an upper limit of) the attenuation coefficient α . From (7a) we get

$$|\Delta\alpha| = \frac{4.34}{L} \frac{|\Delta K|}{K}. \quad (8)$$

The absolute error $|\Delta\alpha|$ of the attenuation coefficient determined by the resonator method is the smaller the higher the contrast and therefore the lower the attenuation coefficient is. This is complementary to most of the usual methods of measuring waveguide losses.

In the discussion above we have assumed a single-frequency light source to have a long coherence length compared with the waveguide length. With axially multimode lasers, as we used as the $\lambda = 1.15 \mu\text{m}$ source, this relation is not fulfilled. This leads to a reduction of the contrast of the Fabry-Perot resonances; an evaluation of the attenuation coefficient, as described above, would yield a value higher than the true one.

However, if the relative intensities of the longitudinal modes are known (they can be measured with a

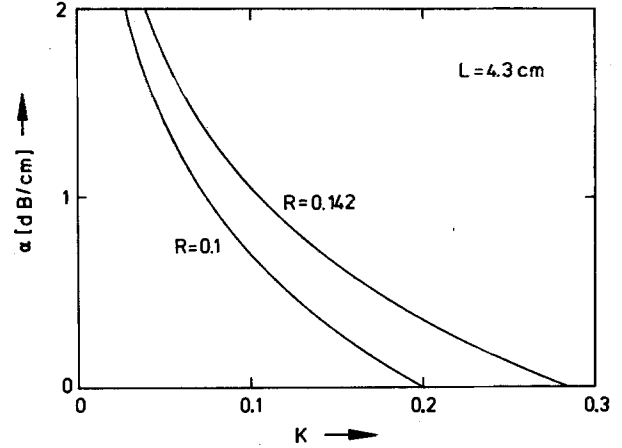


Fig. 4. Waveguide attenuation coefficient α versus the contrast K . Parameters are resonator length L and end face reflectivities R

conventional optical spectrum analyzer) and if these intensity ratios are stable during the period of measurement, the response function of the waveguide resonator is a superposition of its response to the single lines alone. Equation (1) is therefore replaced by a summation of terms which differ only in the wavelength-dependent phases ϕ and intensities I_0 . We presume for this evaluation that the detecting photodiode is slow enough to integrate the difference frequency terms.

Up to this point, we have discussed resonances in monomode waveguide structures. With multimode guides the Fabry-Perot resonator method can be applied as well; the prior condition is that it is possible to excite a distinct mode alone. This, however, is difficult to achieve by end-fire coupling; it turns out that in multimode structures nearly always a combination of modes is excited. As a consequence, a superposition of the response functions of the various waveguide modes would be observed yielding a reduction of the measured contrast. Therefore the most reliable results are obtained with monomode guides, as discussed in the following section.

3. Experimental Results

Figure 5 presents the experimental results obtained with a Ti:LiNbO₃ strip waveguide resonator of 43 mm length on a z-cut crystal. Light propagation was in x-direction. The waveguide had been prepared by indiffusion of a 350 Å thick Ti-stripe of 3 μm width at 1060 °C for 9 h; it was monomode at the wavelength $\lambda = 0.63 \mu\text{m}$. We used a single-frequency He-Ne laser ($\lambda = 0.63 \mu\text{m}$) to study the waveguide resonator. In Fig. 5, the transmitted intensity is plotted versus the crystal temperature. About 13.5 orders of interference are observed per degree Celsius; counting of fringes can therefore be used for temperature measurement.

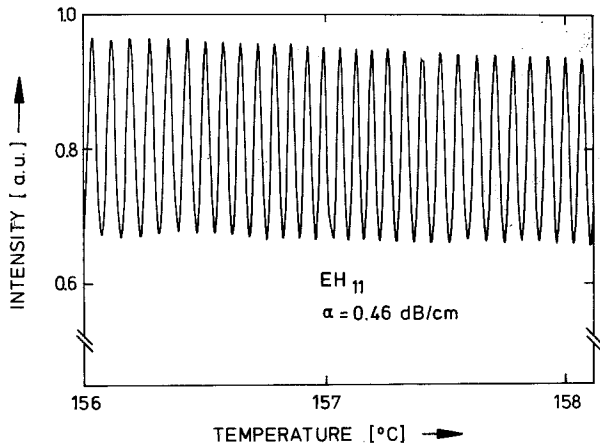


Fig. 5. Transmitted intensity ($\lambda=0.63\ \mu\text{m}$) versus temperature obtained with a $3\ \mu\text{m}$ wide strip waveguide resonator of 43 mm length on a z-cut crystal

The measured contrast K of this example, as defined by (2), is 0.18. Following the discussion of the preceding section, an attenuation coefficient $\alpha=0.46\ \text{dB/cm}$ was evaluated (Fig. 4). The relative accuracy taking a ΔK of 0.01 into account is about 12% leading to $\alpha=0.46\pm0.05\ \text{dB/cm}$. Due to an imperfect fabrication process of the waveguide edges the end-face reflectivities could be lower than the assumed figure given by the modified Fresnel coefficient of (6) ($R=0.142$). In such a case the true attenuation coefficient would be lower than the calculated value of $0.46\ \text{dB/cm}$; this number therefore is an upper limit of the true attenuation coefficient.

Another example is presented in Fig. 6 as a result obtained with a 42.5 mm long waveguide resonator on a y-cut crystal. Light propagation was in x-direction. The sample had been fabricated by diffusion of a $720\ \text{\AA}$ thick Ti-stripe of $10\ \mu\text{m}$ width into LiNbO_3 at $1080\ ^\circ\text{C}$

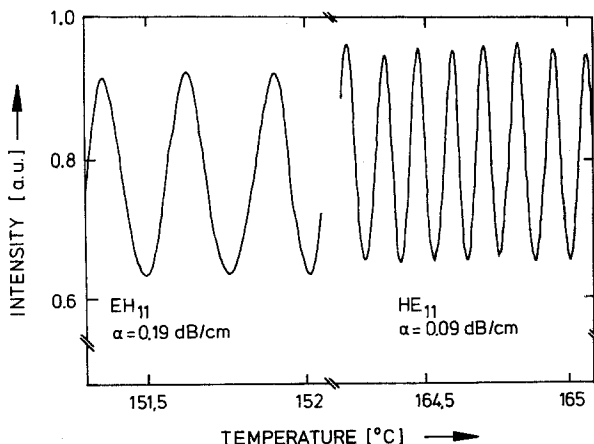


Fig. 6. Transmitted intensity ($\lambda=1.15\ \mu\text{m}$) versus temperature obtained with a $10\ \mu\text{m}$ wide strip waveguide resonator of 42.5 mm length on a y-cut crystal

for 9 h, followed by end-face polishing. The waveguide was monomode at the wavelength $\lambda=1.15\ \mu\text{m}$ for polarization parallel to the surface; with orthogonal polarization up to two lateral modes could be excited.

The investigations were done with a He-Ne laser ($\lambda=1.15\ \mu\text{m}$) having 2–3 longitudinal modes of 385 MHz mode spacing $\Delta\nu$. Therefore the coherence length is reduced to approximately 39 cm ($c/2\Delta\nu$), leading to a reduction of the contrast of the Fabry-Perot resonances [12]. It is evident that this reduction is the more pronounced the longer the waveguide resonator is. To avoid a misinterpretation of the experimental data (the measured contrast K) the evaluation of the attenuation coefficient α (respectively of $\tilde{R}$) has to take the longitudinal spectrum of the laser beam into account, as discussed in Sect. 2. As a result we get the attenuation coefficients $\alpha(\text{EH}_{11})=0.19\pm0.07\ \text{dB/cm}$ and $\alpha(\text{HE}_{11})=0.09\pm0.06\ \text{dB/cm}$. The higher coefficient of the EH_{11} mode could be due to a weak excitation of the EH_{12} mode simultaneously with the EH_{11} mode, thereby reducing the contrast K . $\alpha=0.09\ \text{dB/cm}$ seems to be the lowest attenuation coefficient reported in the literature for Ti:LiNbO_3 waveguides.

[We recall that the HE_{11} -(EH_{11} -) mode is the fundamental mode with the dominant component of its electrical vector parallel (perpendicular) to the waveguide surface.]

4. Conclusions

Measuring the contrast K of the Fabry-Perot resonances in optical waveguide resonators allows to determine the combined loss-reflection factor $\tilde{R}=R\exp(-\alpha L)$. If the end-face reflectivities for the investigated mode are known, the waveguide attenuation coefficient can be evaluated. The error of this method is the smaller, the lower the waveguide losses are.

On the other hand, unknown mode reflectivities can be determined if the attenuation coefficient is known from another independent measurement. (This could be done by one of the methods cited in the introduction.) Indeed, it is very desirable to measure an end-face reflectivity independently from other parameters: the insertion loss of all integrated optical devices is thereby influenced.

The contrast K respectively the factor $\tilde{R}$ is the ideal parameter to characterize the quality of (low-finesse) optical waveguide resonators. It allows to select good samples with low losses and perfect end-faces before they are coated by dielectric mirrors to fabricate high-finesse resonant devices. Examples are integrated optical spectrum analyzers, resonant second harmonic

generators, and integrated optical parametric oscillators [10].

Furthermore, low- and high-finesse integrated optical resonators are very attractive to study effects of optical damage with high sensitivities. We shall discuss these experiments elsewhere.

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